



DARK PROTEOME SUMMER SCHOOL 2026

Program Booklet

24 – 26 September 2026
RELC International Hotel
30 Orange Grove Road, Singapore 258352

Welcome

It is our great pleasure to welcome you to the inaugural Dark Proteome Summer School 2026 on “**Bridging the gap: RNA translation to hidden proteomes**”.

This is a first-of-its kind RNA Translation and Proteomics summer school with a focus on the novel RNA translation events leading to proteins and aberrant peptide production in cells. Over the next three days, we will bring together researchers, students, and thought leaders to discuss specialized workflows and research in the field. Through keynote lectures, invited talks, and selected flash and short talks from our participants, we hope this summer school will spark new ideas, collaborations, and friendships that extend well beyond these three days.

We are especially grateful to our speakers for their support and sharing their expertise, and to all participants who submitted abstracts and will be presenting their work during the flash and short talk sessions. We would also like to thank our sponsors – Immagina Biotechnology and EMBO - whose generous support made this event possible.

We hope you leave the event with new insights, connections, and inspirations for opportunities ahead in the field.

The Dark proteome summer school 2026 chairs:

*Sonia Chothani, Genome Institute of Singapore, A*STAR*

Marie A. Brunet, University of Sherbrooke, Canada

Meeting coordinator

*Devika Menon, Genome Institute of Singapore, A*STAR*

Program Schedule — Day 1

[Thursday, 24 September 2026]

12:15 pm – 1:15 pm

Registration

1:30 pm – 3:30 pm

S01: Foundation of RNA Translation & Ribo-seq [Room 502, Level 5]

- 1:30 pm – 2:10 pm: **THE EMBO KEYNOTE LECTURE** and demo by **Pavel Baranov** (University College Cork, Ireland)
Foundations of RNA Translation and the Expanding Coding Landscape
- 2:10 pm – 2:50 pm: Talk and Demo by **Eivind Valen** (University of Oslo, Norway)
Massive aggregation and stratification of ribosome profiling reveals cryptic translation and regulatory complexity
- 2:50 pm – 3:20 pm: Talk by **Zhi Xie** (Guangzhou National Laboratory, China)
Learning to Predict and Engineer mRNA Translation from Large-Scale Ribo-seq Data.
- 3:20 pm – 3:30 pm: Sponsor Talk by **Massimiliano Clamer**, Immagina Biotech, *Technology Solutions for Translation Profiling and Ribosome-associated Workflows*

03:30 pm – 4:00 pm

Coffee Break

4:00 pm – 5:30 pm

S02: Experimental Workflows and Pre-processing [Room 502, Level 5]

- 4:00 pm – 5:00 pm: Talk and Hands-on Demo by **Sonia Chothani** (Genome Institute of Singapore, A*STAR, Singapore)
Pre-processing, ORF Detection and differential translation analysis
- 5:00 pm – 5:15 pm: Selected short talk by **Sandeep M Eswarappa**
Leaky scanning translates a conserved ORF in the 3'UTR of RPL36A and regulates the expression of ribosomal protein L36a
- 5:15 pm – 5:30 pm: Selected short talk by **Hongyan Wang**
Microproteins Simba1 and Simba2 activate Wingless signaling during the reactivation of neural stem cells in Drosophila

6:00 pm

Gather at L1 lobby RELC hotel @5:45 pm for Singapore sightseeing / activity

Program Schedule — Day 2

[Friday, 25 September 2026]

9:00 am – 10:30 am

S03: Applications and Other Ribosome-associated Technologies [Room 502, Level 5]

- 9:00 am – 9:30 am: Talk by **Yoshika Janapala** (The John Curtin School of Medical Research),
Decoding the rapid translational response to glucose starvation using eTCP-seq.
- 9:30am – 9:45am: Selected short talk by **Félix-Antoine Trifiro**
OpenProt 3.0 enables robust discovery of non-canonical proteins through class-specific FDR and modification-aware proteomics
- 9:45 am – 10:30 am: Selected flash talks
Jose Antonio Flores - *Invisible Players in Antiviral Immunity: Exploring Microprotein Regulation*
Runguo Shu - *Transcriptome-wide Profiling of Evolutionary Constraints on Human smORF-Encoded Peptides*
Jiapeng He - *sORF-Trans2MS: a two-module deep learning framework for sORF translation and MS-supported microprotein prediction*
Isabelle Perron-Lepine - *Functional resurrection of a GAPDH pseudogene: a peptidein controlling mitochondrial cristae architecture*
Dinglan WU - *Dynamic Profiling of Microproteins During the Evolution of Androgen Deprivation Resistance in Prostate Cancer*

10:30 am – 11:00 am

Coffee Break

11:00 am – 12:00 pm

S04: Foundations and outlook of proteomics and immunopeptidomics [Room 502, Level 5]

- 11:00 am – 11:30 am: Talk by **Lennart Martens** (Ghent University & VIB, Belgium)
Making the bizarre mundane: searching for non-canonical proteoforms at scale
- 11:30 am – 12:00 noon: Talk by **Wu Wei** (Singapore Immunology Network, A*STAR, Singapore)
Connecting non-classical neoantigen discovery with non-canonical translation

12:00 pm – 1:00 pm

Lunch

01:00 pm – 03:30 pm

S05: Proteomic Workflows for Non-canonical Protein Identification [Room 502, Level 5]

- 1:00 pm – 1:45 pm: Lecture and demo by **Eric Deutsch** (Institute for Systems Biology, Seattle, USA)
Validating Peptidein Detections with PeptideAtlas
- 1:45 pm – 2:45 pm: Talk and hands-on-demo by **Marie Brunet** (Sherbrooke University, Canada)
Considerations for robust identification of peptideins; Targeted Proteomics for Rare Peptide Discovery and Quantitative Protein Dysregulation
- 2:45 pm – 3:00 pm: Selected short talk by **Chris Soon Heng TAN**
Combining Orthogonal Enrichment Methods to Maximize MS-based Microprotein Discovery
- 3:00 pm – 3:30 pm: Panel discussion: **Spectral Validation for Confident Identification**. What Evidence Is Enough for Confident Identification of smaller or non-canonical proteins?

3:30 pm – 4:00 pm

Coffee Break

4:00 pm – 5:00 pm

S06: Applications and Thoughts for the future [Room 502, Level 5]

- 4:00 pm – 4:15 pm: Selected short talk by **Riko Egawa**
Comprehensive Identification of Noncanonical Translation Products Using Multi-Protease Terminomics and Expanded Translation Databases
- 4:15 pm – 4:30 pm: Selected short talk by **Qian Zhao**
Engineering the Microprotein Workflow: From Enrichment, TMT-labelling, Acquisition to DIA Libraries
- 4:30 pm – 5:00 pm: Panel Discussion. Thoughts for the future. What are the main challenges our field is facing and what should we tackle in the next 5 years?

Program Schedule — Day 3

Combined session with HUPO-HIPP summer school

[Saturday, 26 September 2026]

09:00 am – 12:00 pm

S07: Multi-omic discovery and applications [Room 503-504, Level 5]

- 9:00 am – 9:30 am: Talk by **Jenn Abelin** (BROAD institute and Dana-Farber Cancer Institute, USA)
Beyond the Canonical Proteome: Immuno-peptidomics as a Tool for Dark Proteome Discovery
- 9:30 am – 10:00 am: Talk by **Yardena Samuels** (Weizmann Institute of Science, Israel)
Challenges and Solutions in Effectively Targeting Neoantigens
- 10:00 am: **Dark proteome summer school 2026 group photo.**

10:00 am – 10:30 am

Coffee Break

- 10:30 am – 11:00 am: Talk by **Sarah Slavoff** (Yale Institute of Biomolecular Design and Discovery)
Quantitative, Chemical and Functional Proteomics of Microproteins
- 11:00 am – 11:30 am: Talk by **Michal Bassani-Sternberg** (Ludwig Institute for Cancer Research / UNIL-CHUV, Lausanne, EU)
Multi-omic Discovery of Non-canonical Immuno-peptidome Sources: Challenges and Emerging Opportunities
- 11:30 am – 12:00 pm: Panel Discussion and Closing

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